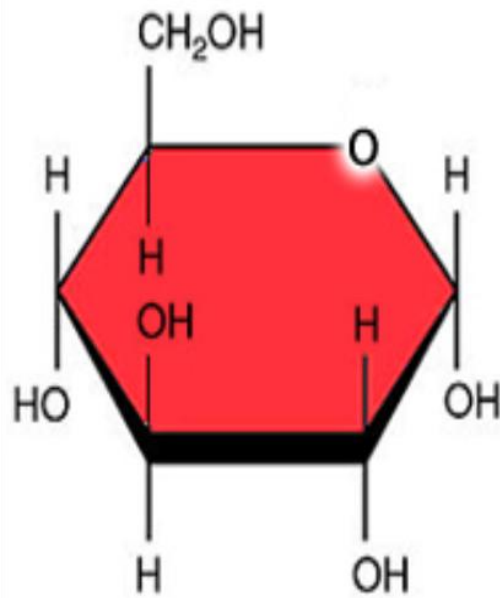
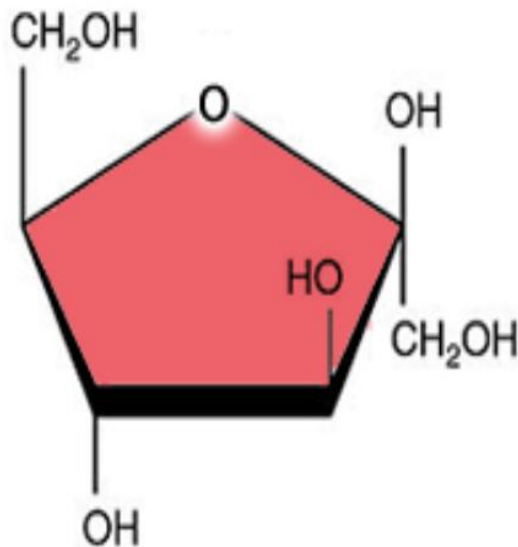


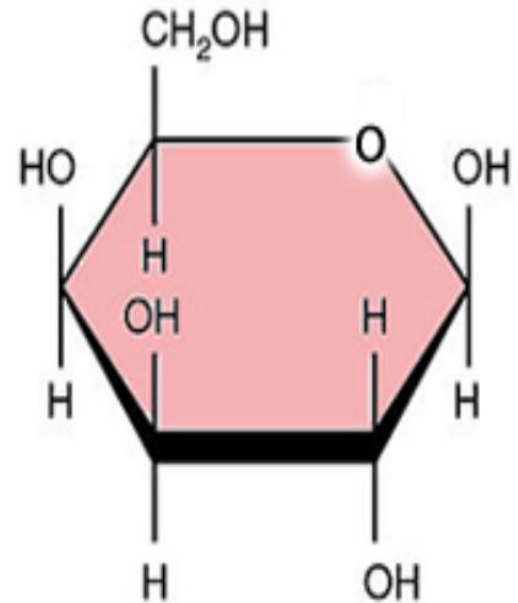
Monosaccharides



Glucose



Fructose



Galactose

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Monosaccharides

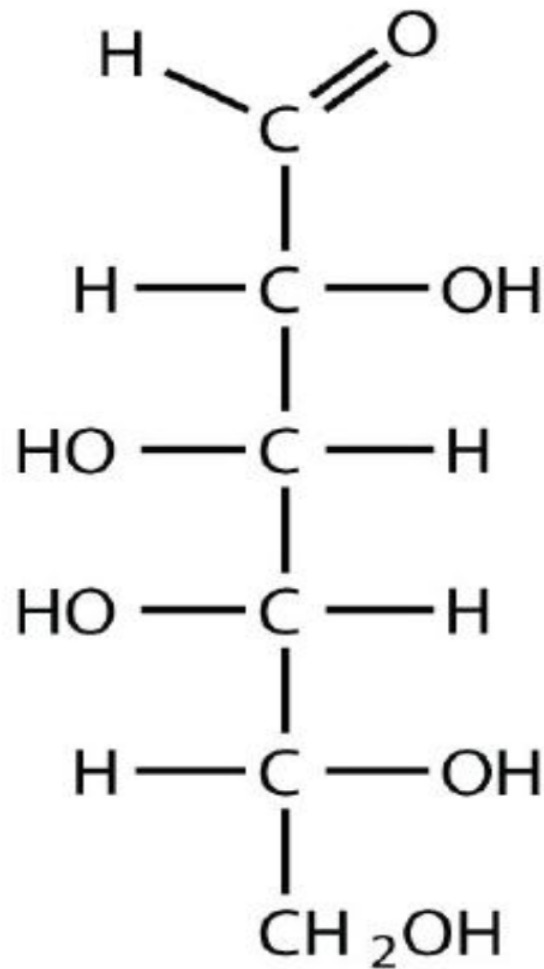
The most frequent monosaccharides are the hexose sugars; **glucose**, **fructose** and **galactose**.

HEXOSE SUGARS

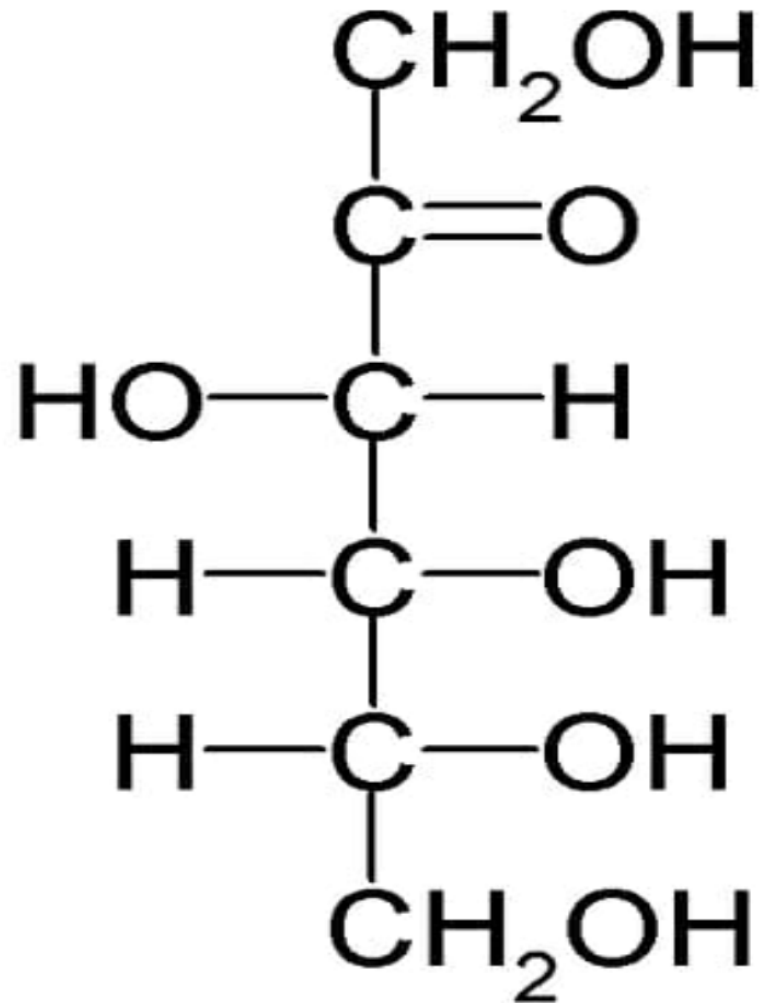
These are sugars with molecular formula $\text{C}_6\text{H}_{12}\text{O}_6$ and structural formulae as shown in the next slide:

Glucose (aldose)

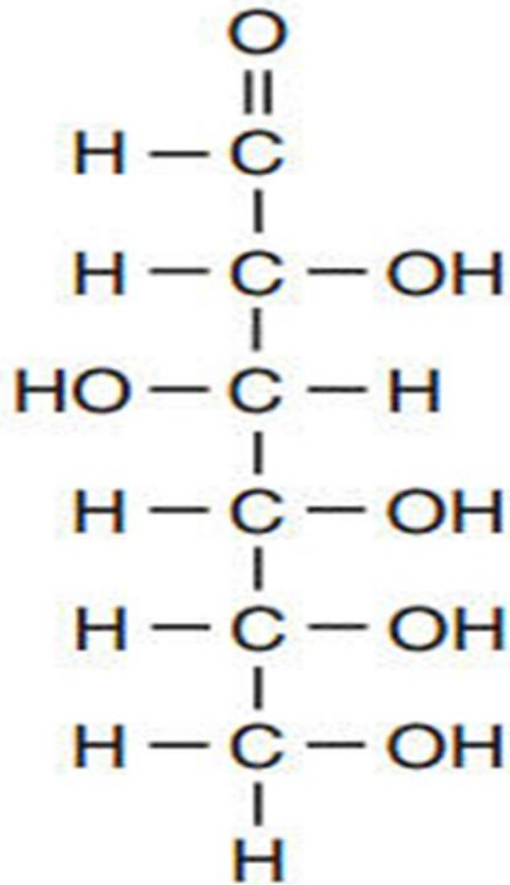
Aldose sugar



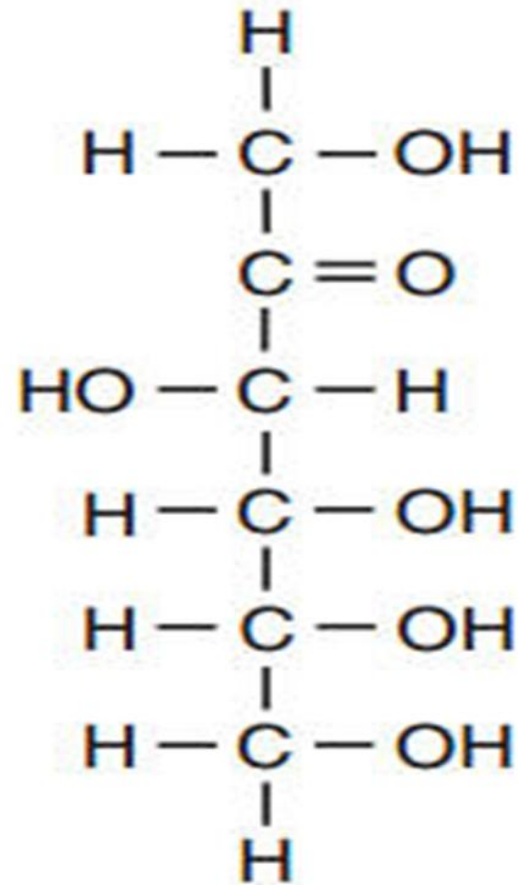
Ketose sugar



Aldose or Ketose sugar



Glucose

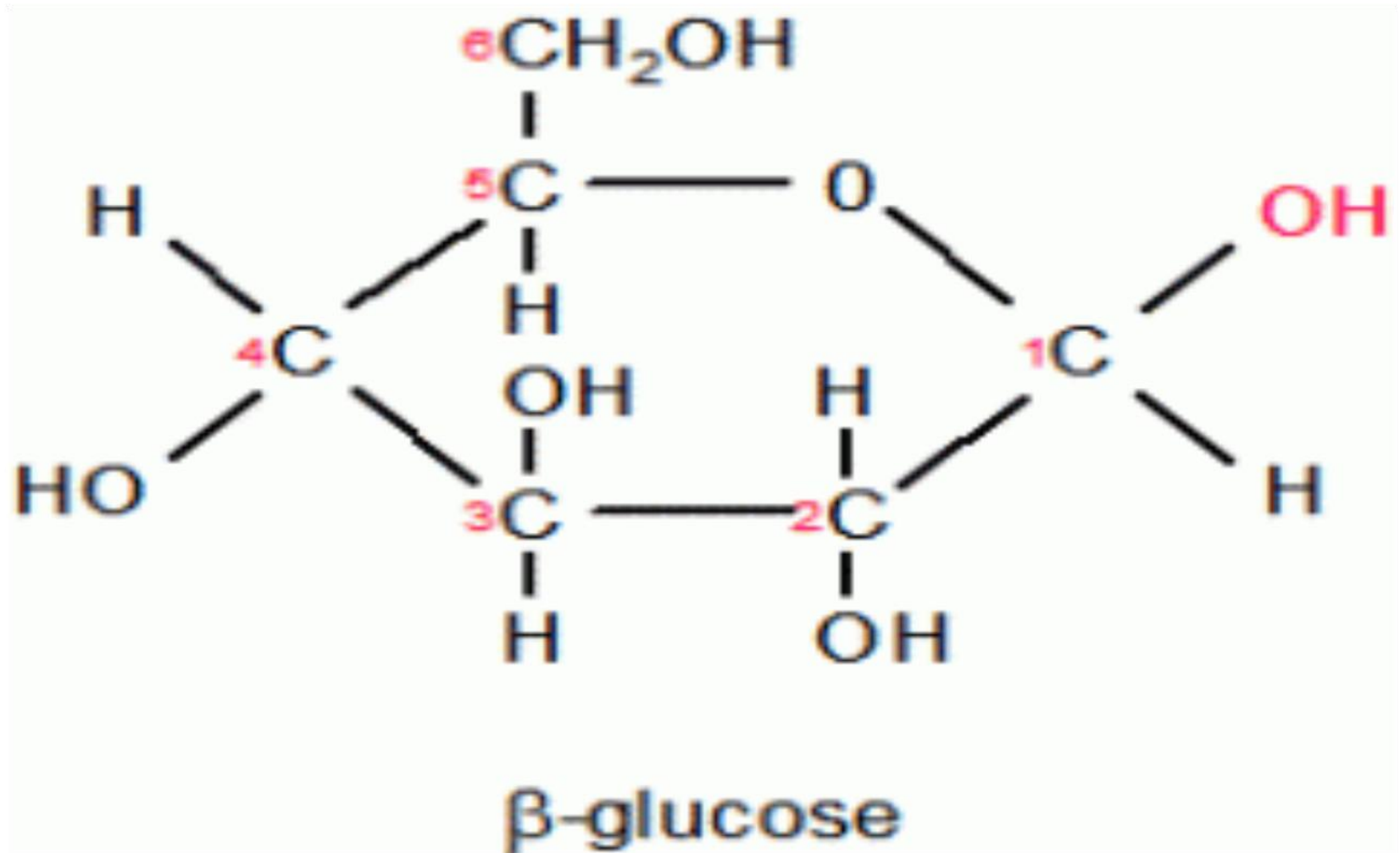


Fructose

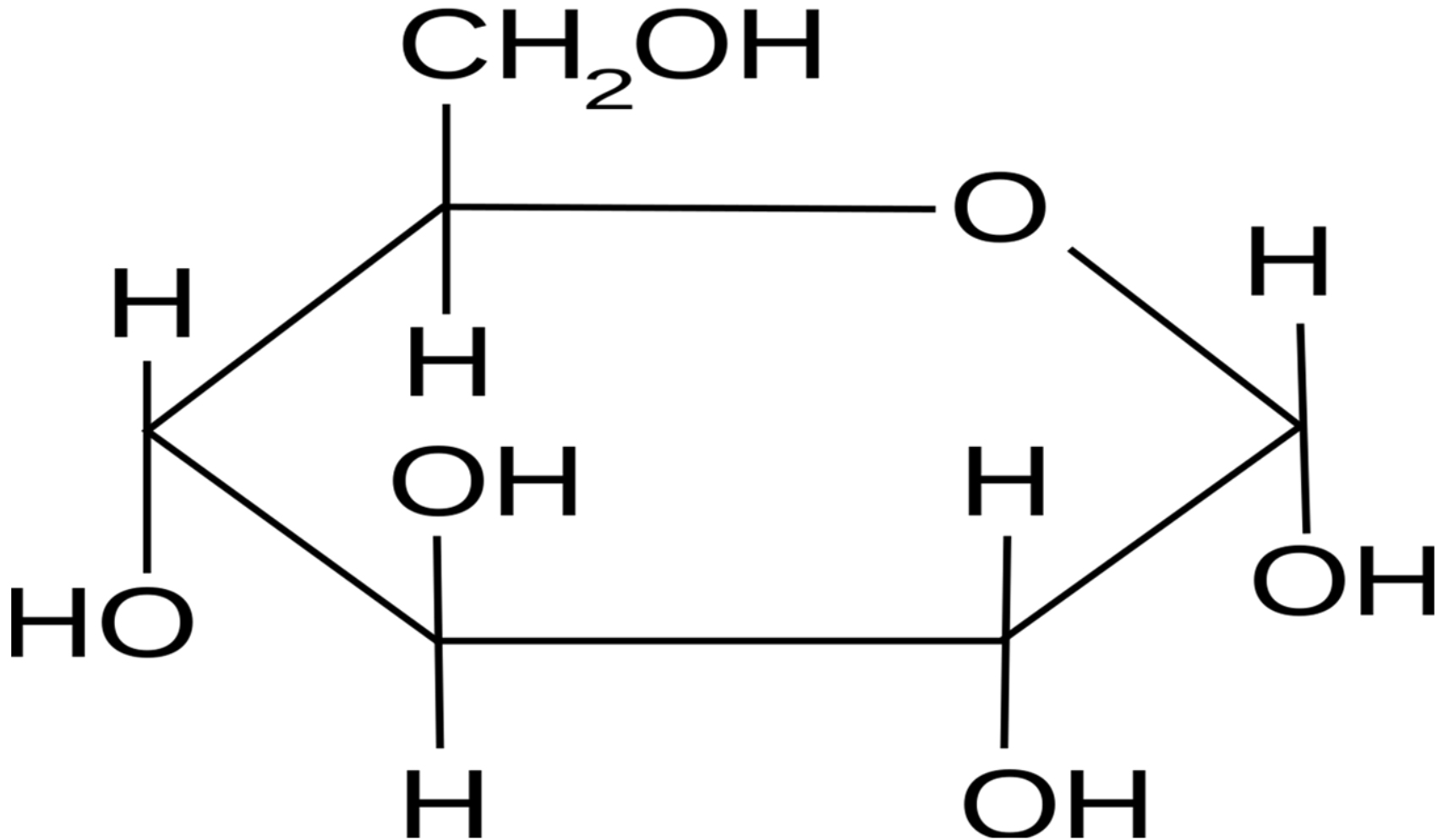
Isomerism in monosaccharides

Glucose can exist in a number of isomers where it has *different structural formulae* but the same molecular formulae.

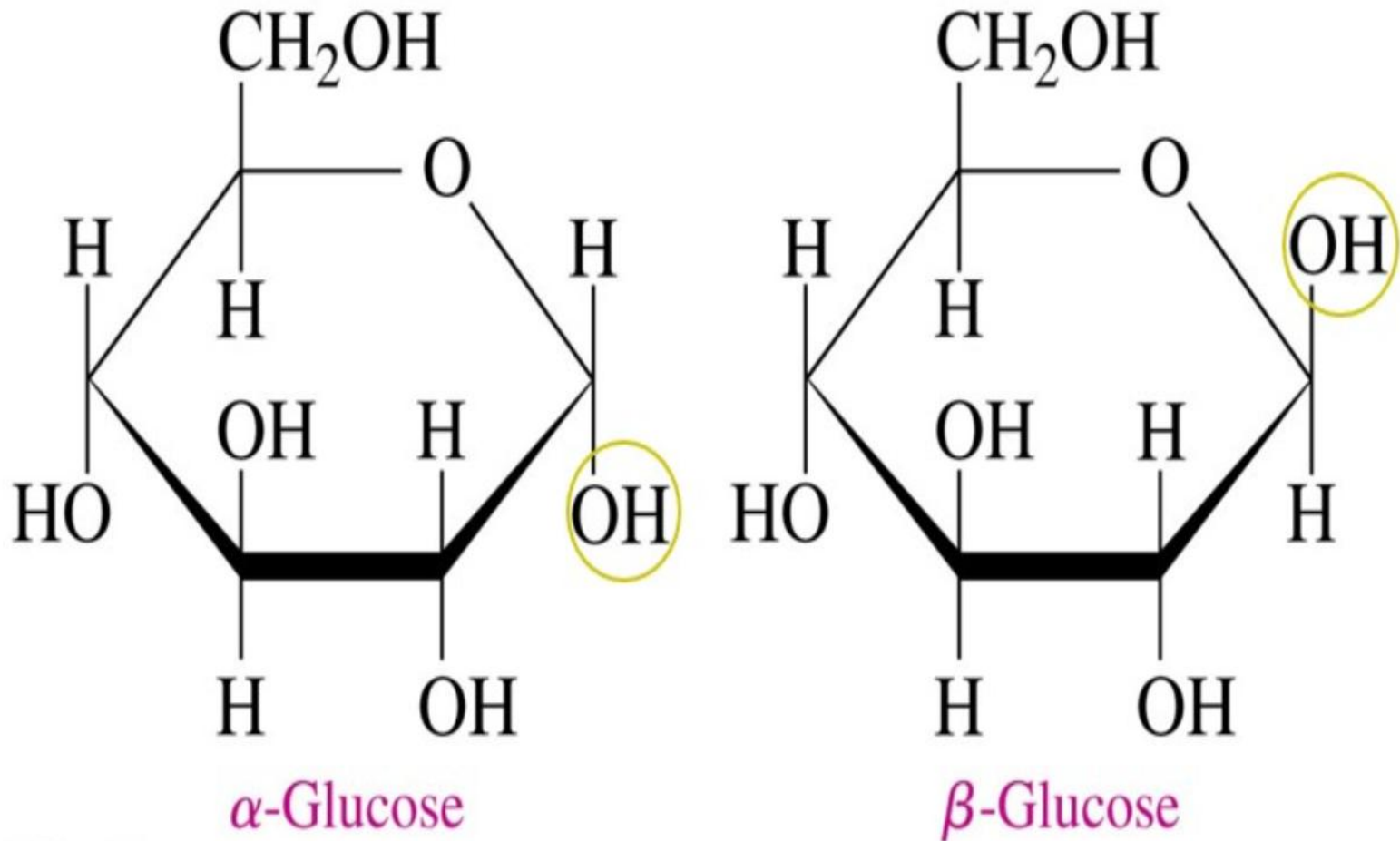
Isomers of glucose (beta glucose)



Isomers of glucose (Alpha glucose)



Alpa and Beta isomers



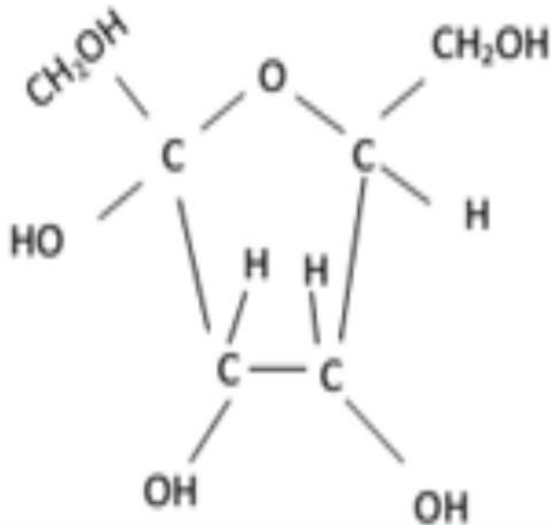
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Structures of hexoses

The hexose sugars can exist in *straight/chain* form as shown above or in *ring form* as shown in the next slide.

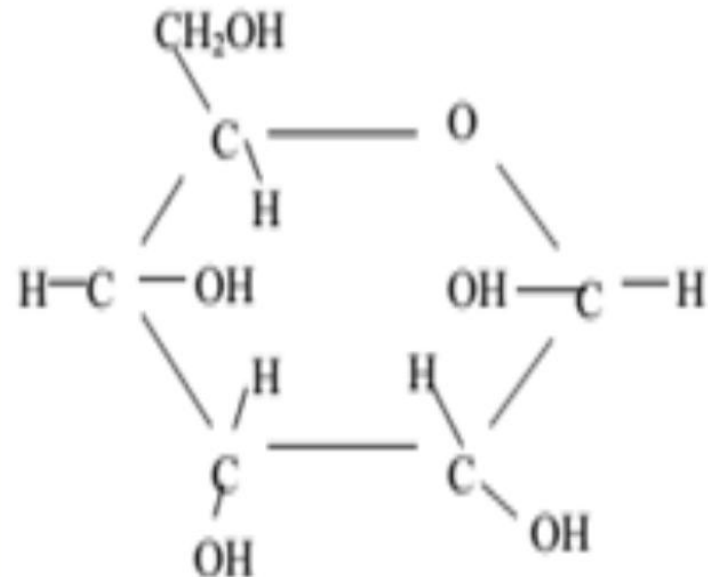
STRUCTURES OF HEXOSE SUGARS

Fructose



FRUCTOSE STRUCTURE

Glucose



STRUCTURE OF GLUCOSE

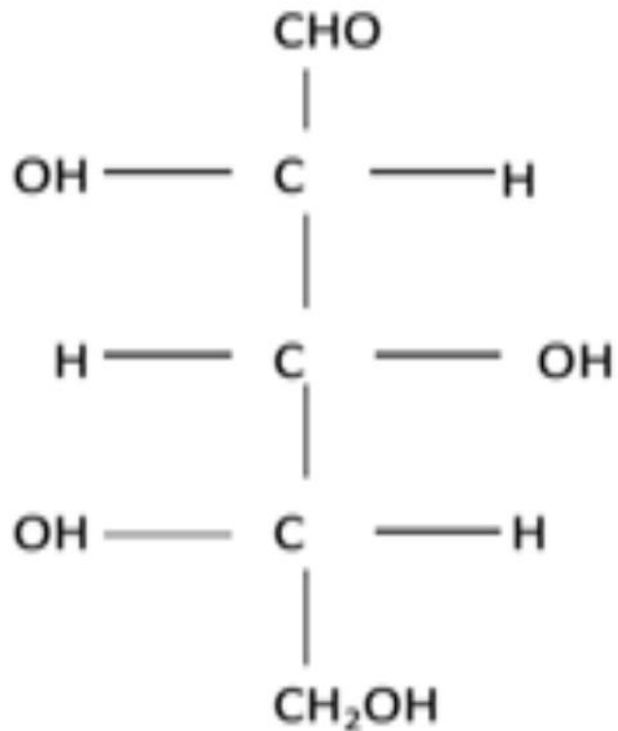
PENTOSE SUGARS

They have **5 carbon atoms**.

They are found in nature as **ribose** and **deoxyribose**.

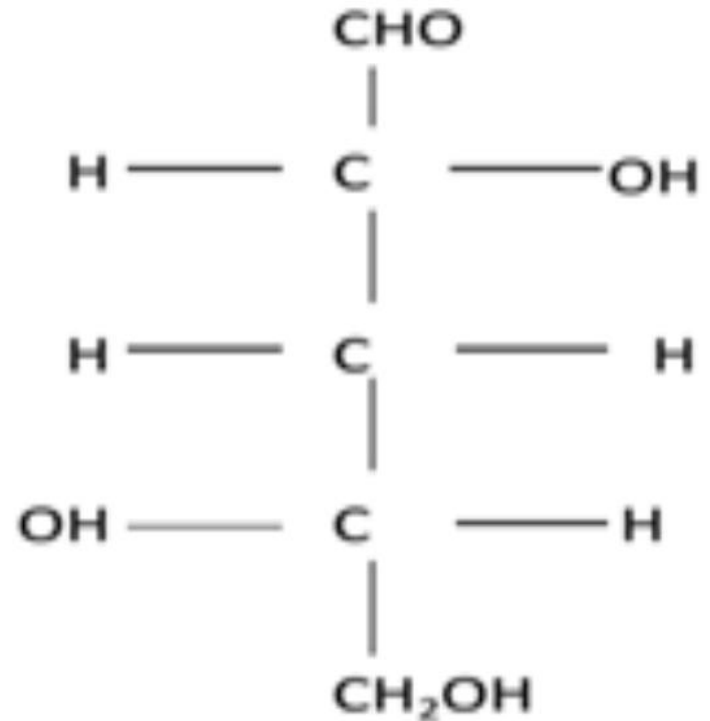
They exist in *straight* and *ring* forms.

Ribose



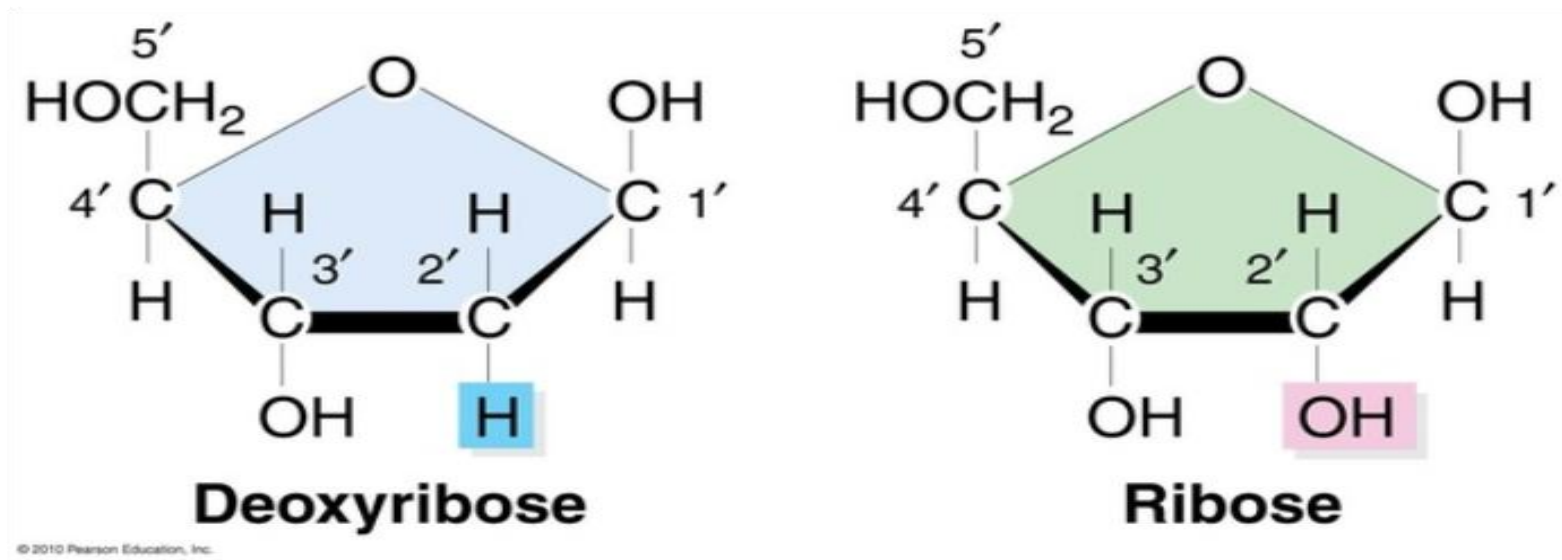
RIBOSE

Deoxyribose



DEOXYRIBOSE

Ring forms of hexose sugars



Ribose occurs in co-enzymes, adenosine triphosphate (**ATP**) and Ribonucleic acid (RNA). Deoxyribose occurs in DNA (*Deoxyribo Nucleic Acid*).

TRIOSE SUGARS

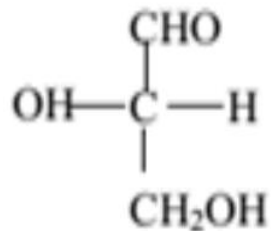
They contain **3 carbon atoms**.

The two occurring trioses are **glyceraldehyde** and **dehydroxyacetone**.

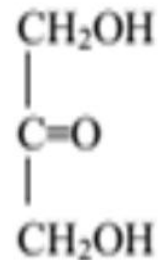
Both of them are found in ***plant*** and ***animal*** cells playing a role in carbohydrate metabolism

Triose sugar structures

Glyceraldehyde

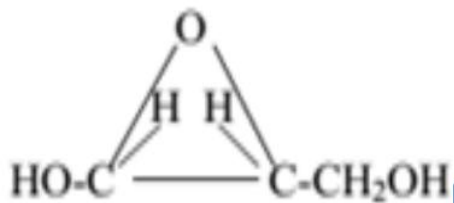


Dehydroxyacetone

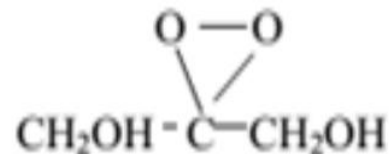


Ringed forms

Glyceraldehyde

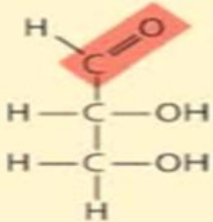
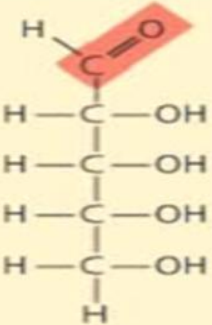
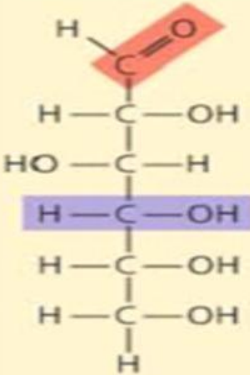
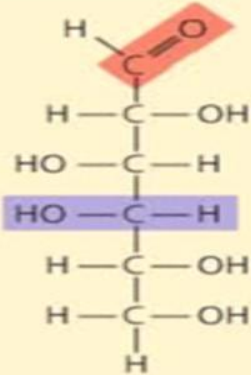
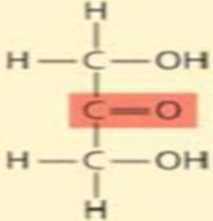
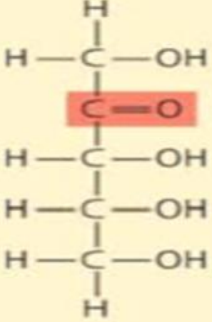
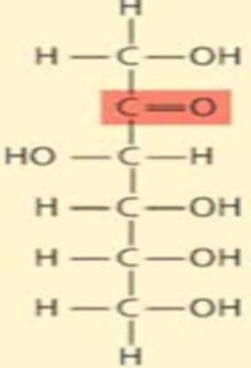


Dehydroxyacetone



TRIOSE SUGARS

MONOSACCHARIDES SUMMARY

	Trioses ($C_3H_6O_3$)	Pentoses ($C_5H_{10}O_5$)	Hexoses ($C_6H_{12}O_6$)	
Aldoses	 Glyceraldehyde An initial breakdown product of glucose	 Ribose A component of RNA	 Glucose An energy source for organisms	 Galactose An energy source for organisms
Ketoses	 Dihydroxyacetone An initial breakdown product of glucose	 Ribulose An intermediate in photosynthesis	 Fructose An energy source for organisms	

DISACCHARIDES

Monosaccharides combine together in pairs to form **disaccharides**. This union involves loss of a water molecule and therefore the reaction is a condensation reaction.

The bond formed is a **glycosidic bond**. The most common disaccharides are:

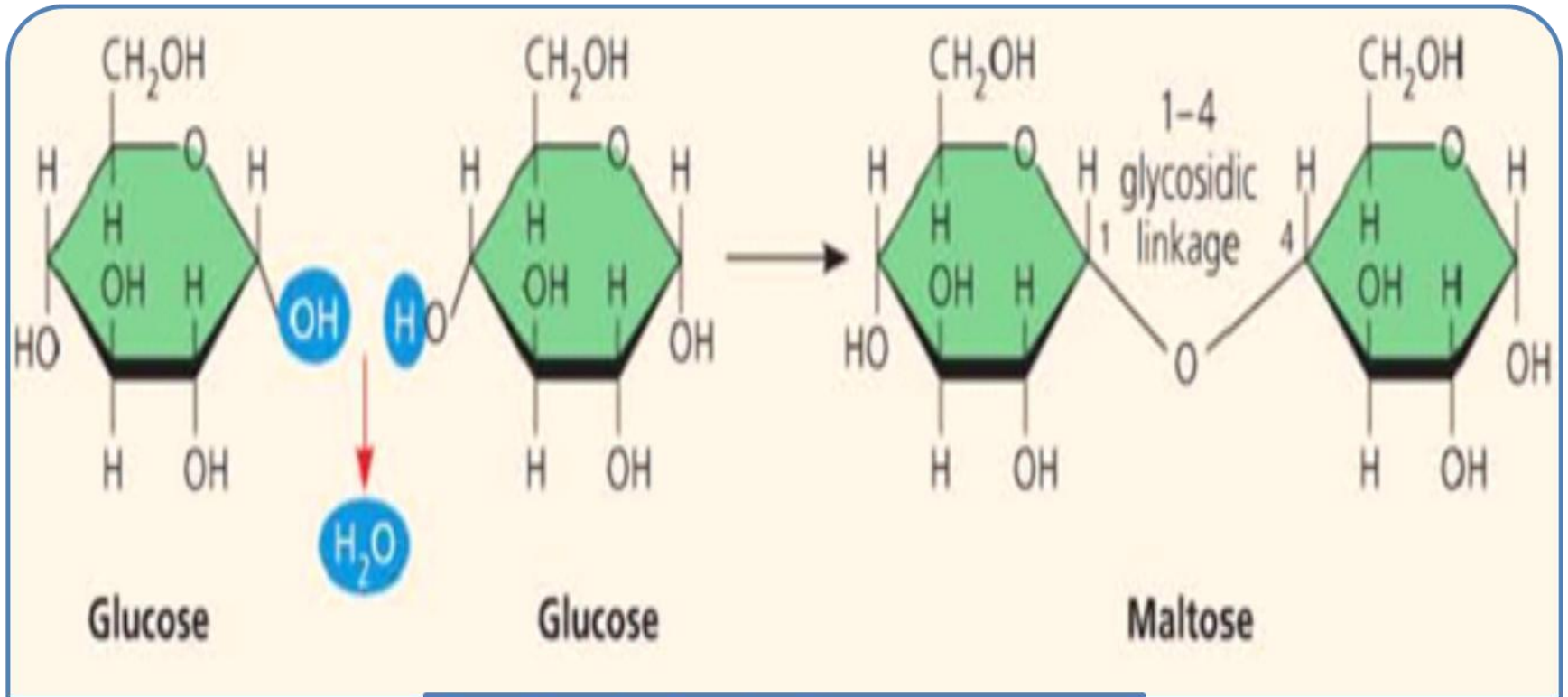
- ❖ **Maltose** formed from linkage of two glucose molecules. It is common in most ***germinating seeds*** and cereals.
- ❖ **Sucrose** from union of glucose and fructose. It is the main form in stems of **sugar canes** and roots of sugar beets which are sources of commercial sugars.
- ❖ **Lactose** resulting from the union of glucose and galactose and found in **milk**.

The disaccharides have the following properties:

- ❖ They are sweeter than monosaccharides
- ❖ They can be crystallized
- ❖ They are soluble in water
- ❖ Do not change the colour of **Benedict's solution** when heated with it (apart from maltose)- they are known as non-reducing sugars
- ❖ Can be broken down into simple sugars by dilute mineral acids and enzymes

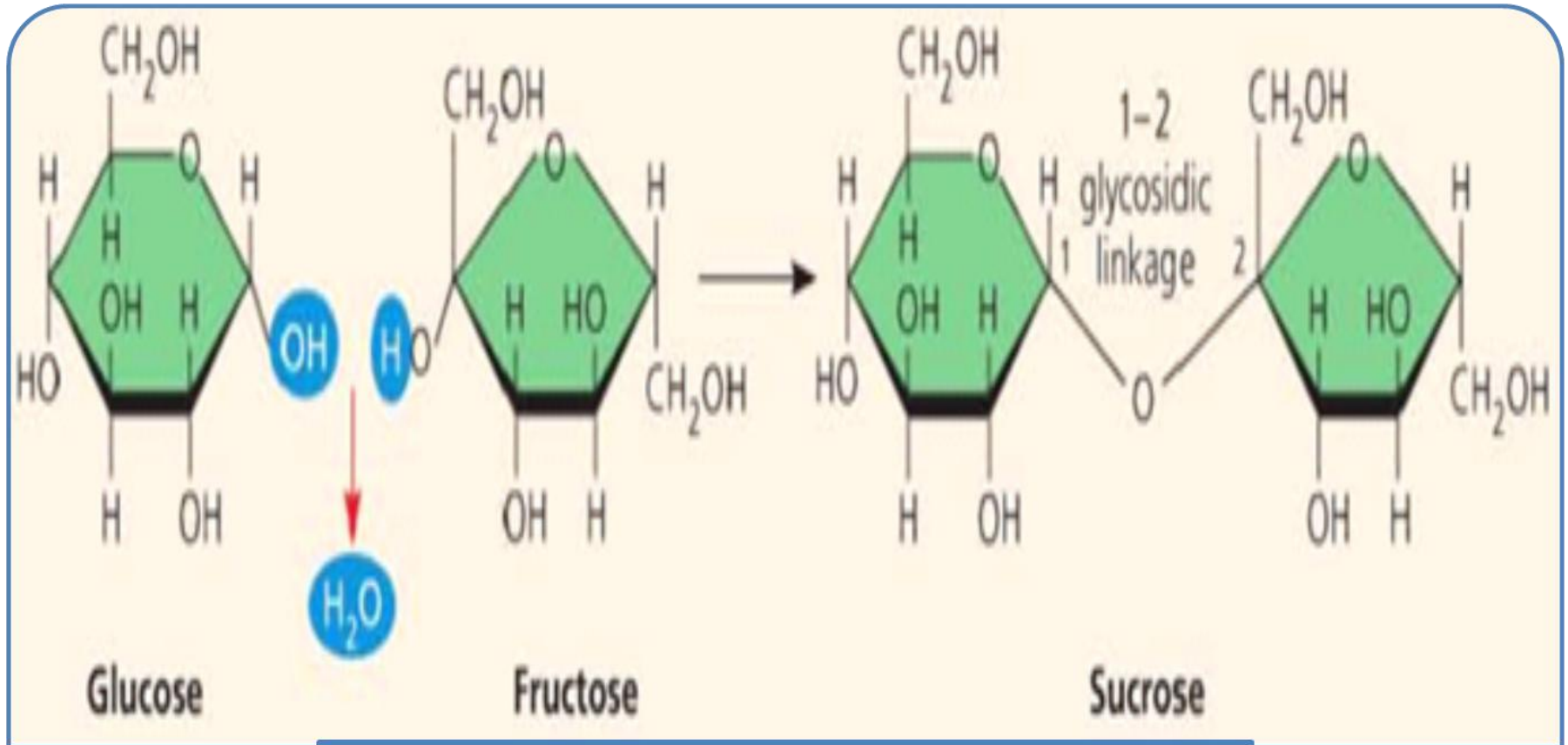
Note: Maltose can also be formed as a product of starch hydrolysis

Formation of maltose



Formation of a disaccharide –
maltose

Formation of sucrose



Formation of sucrose